

- (ii) Die y-komponente van die kragte is.
The y-components of the forces are.

$$\sum F_y = -mg + F_N \cos \theta$$

$$= -mg + \left(\frac{mg}{\cos \theta} \right) \cos \theta = 0$$

(1)

- (iii) Toon aan dat die speed van die kar gegee word deur $v = (gR \tan \theta)^{\frac{1}{2}}$.

Show that the speed the driver should take the curve is given by $v = (gR \tan \theta)^{\frac{1}{2}}$.

$$\therefore \frac{mg \sin \theta}{\cos \theta} = m \frac{v^2}{R}$$

$$\therefore (\tan \theta) g R = v^2$$

$$\therefore v = (gR \tan \theta)^{\frac{1}{2}}$$

(3)

- (iv) Met $R = 320$ m en $\theta = 5,1^\circ$, bepaal die speed van die voertuig (in km h^{-1}) sodat die voertuig nie afgly nie.

With $R = 320$ m and $\theta = 5,1^\circ$, determine the speed (in km h^{-1}) the driver should take the curve to avoid sliding off the road.

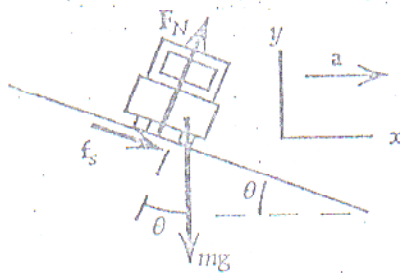
$$v = (gR \tan \theta)^{\frac{1}{2}} = \sqrt{(9,8)(320)(\tan 5,1^\circ)}$$

$$v = 16,73 \text{ m/s} = 60,23 \text{ km/h}$$

(3)

- (v) Nadat die sneeu gesmelt het is die statiese wrywingskoëffisiënt tussen die bande en die pad 0,8.

After the snow melted, the coefficient of static friction between the tyres and the road is 0,8.



x-komponente / x-components

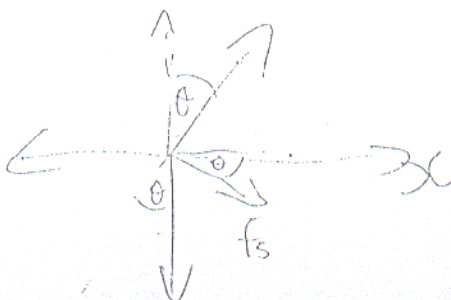
$$\sum F_x = f_s \cos \theta + F_N \sin \theta = m \frac{v^2}{R}$$

(1)

y-komponente / y-components

$$\sum F_y = f_s \sin \theta - mg + F_N \cos \theta = 0$$

(1)



Vraag / Question 2(c)(vi)(Vervolg / Continue) / ...